

VOLTAGE AND CURRENT DIVIDER THEOREM

CIRCUIT ANALYSIS METHOD

prepared by:

Gyro A. Madrona

Electronics Engineer

TOPIC OUTLINE

Voltage Divider Theorem (VDT)

Current Divider Theorem (IDT)



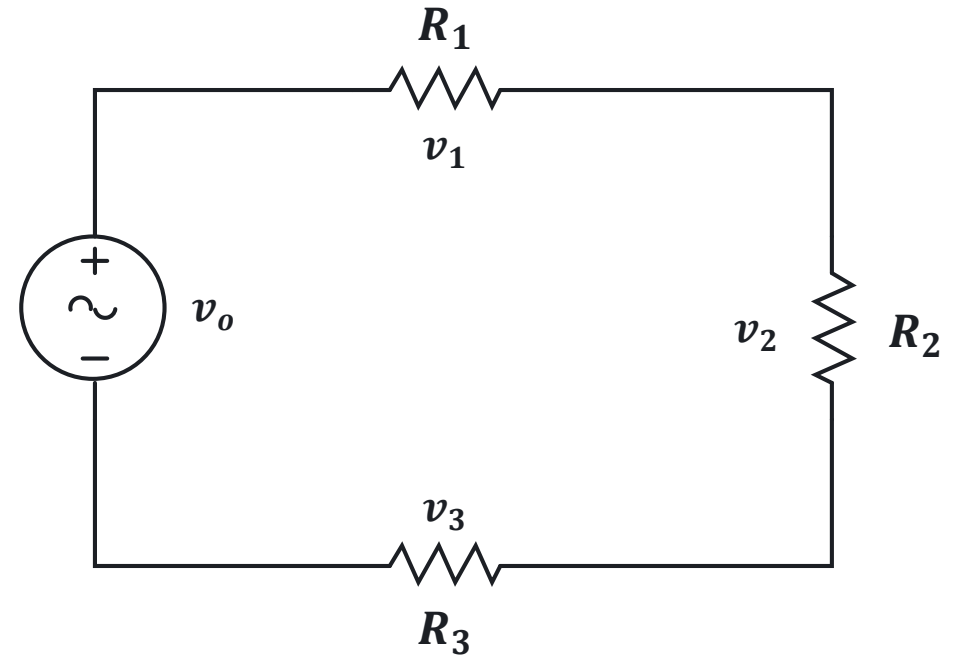
VOLTAGE DIVIDER THEOREM



VOLTAGE DIVIDER THEOREM

A voltage divider consists of two or more resistors connected in **series** with a **voltage source**. The voltage across each resistor is a fraction of the total applied voltage, determined by the ratio of that resistor's value to the total resistance in the series.

Series Network



VOLTAGE DIVIDER THEOREM

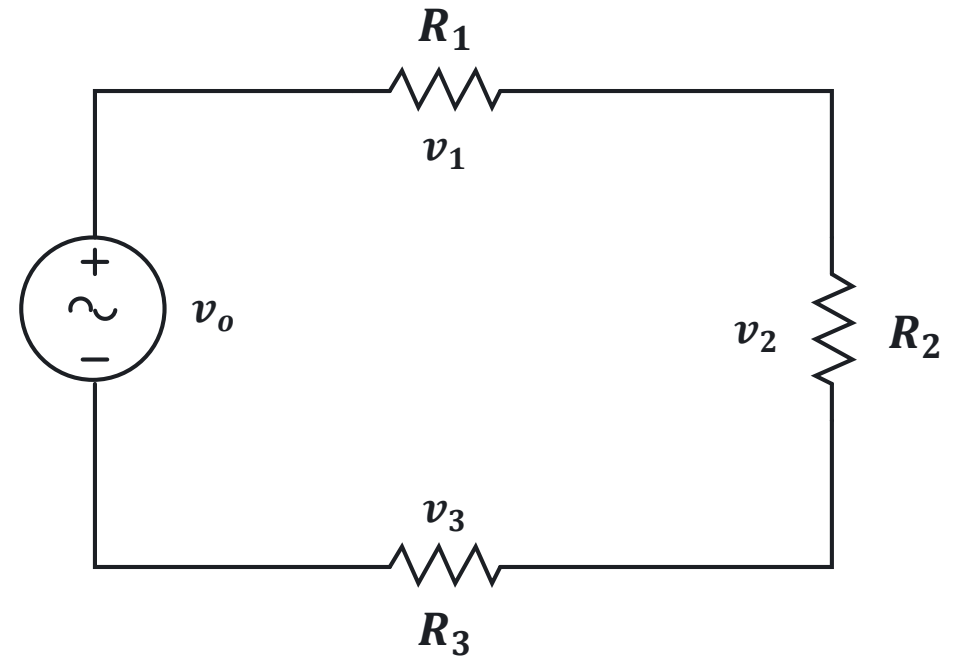
Series Network

$$v_1 = v_o \frac{R_1}{R_o}$$

$$v_2 = v_o \frac{R_2}{R_o}$$

$$v_3 = v_o \frac{R_3}{R_o}$$

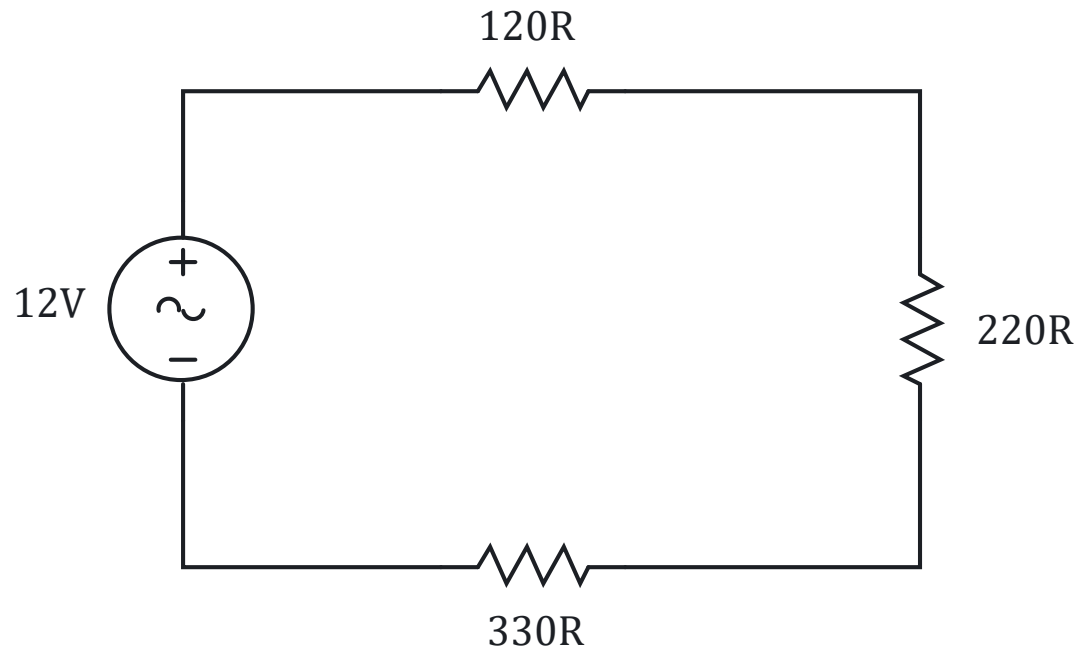
$$v_n = v_o \frac{R_n}{R_o}$$



EXERCISE

Determine the voltage drop across each resistor of the given circuit.

Solution



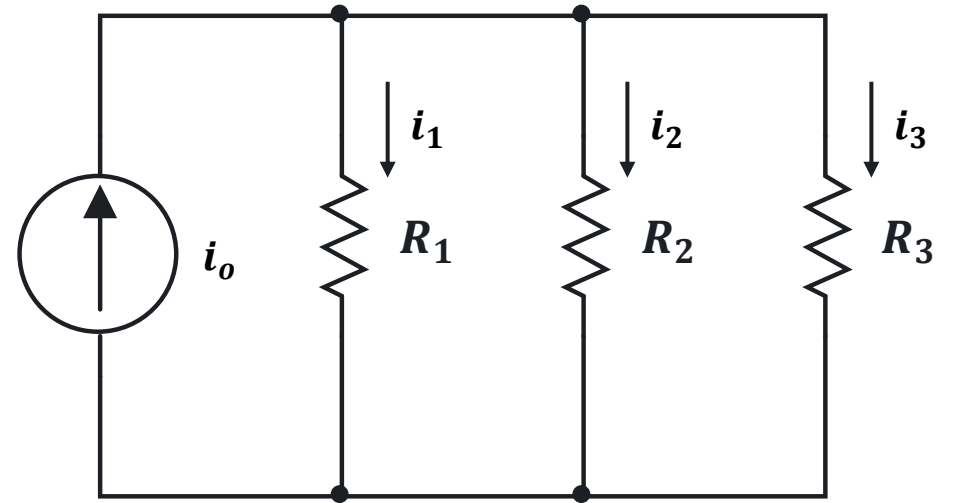
CURRENT DIVIDER THEOREM



CURRENT DIVIDER THEOREM

A current divider is consists of two or more resistors connected in **parallel** with a **current source**. The current through each resistor is a fraction of the total current, determined by the ratio of the other resistor's value to the total resistance in the parallel combination.

Parallel Network



CURRENT DIVIDER THEOREM

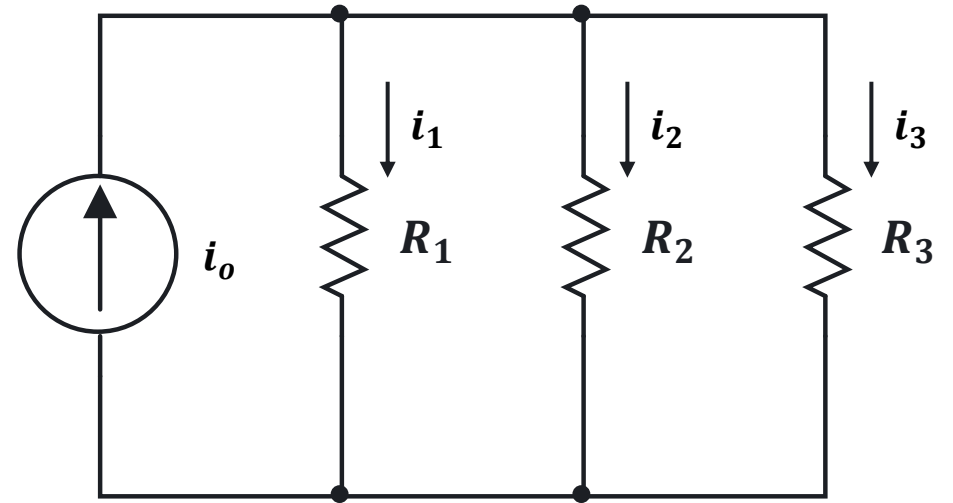
Parallel Network

$$i_1 = i_o \frac{R_o}{R_1}$$

$$i_2 = i_o \frac{R_o}{R_2}$$

$$i_3 = i_o \frac{R_o}{R_3}$$

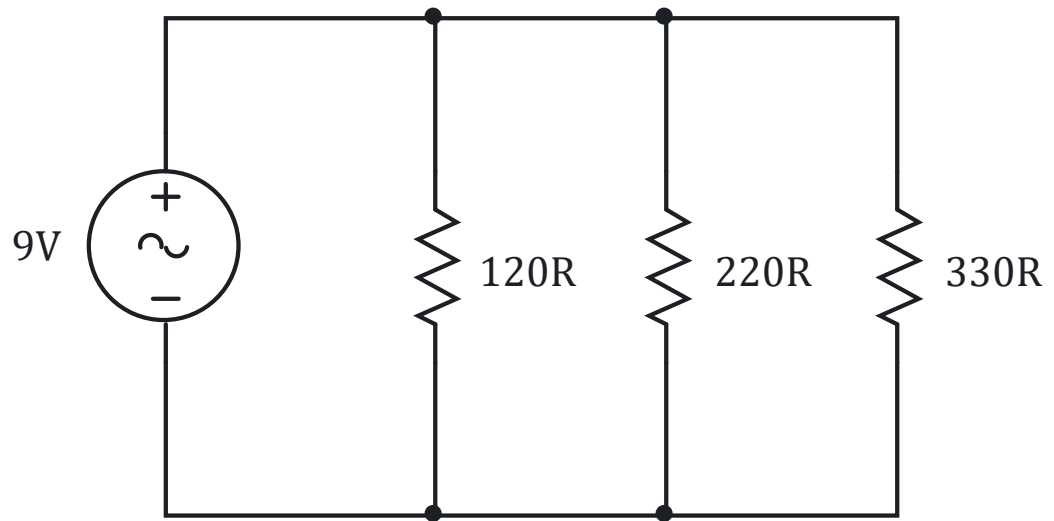
$$i_n = i_o \frac{R_o}{R_n}$$



EXERCISE

Determine the current flowing through each resistor of the given circuit.

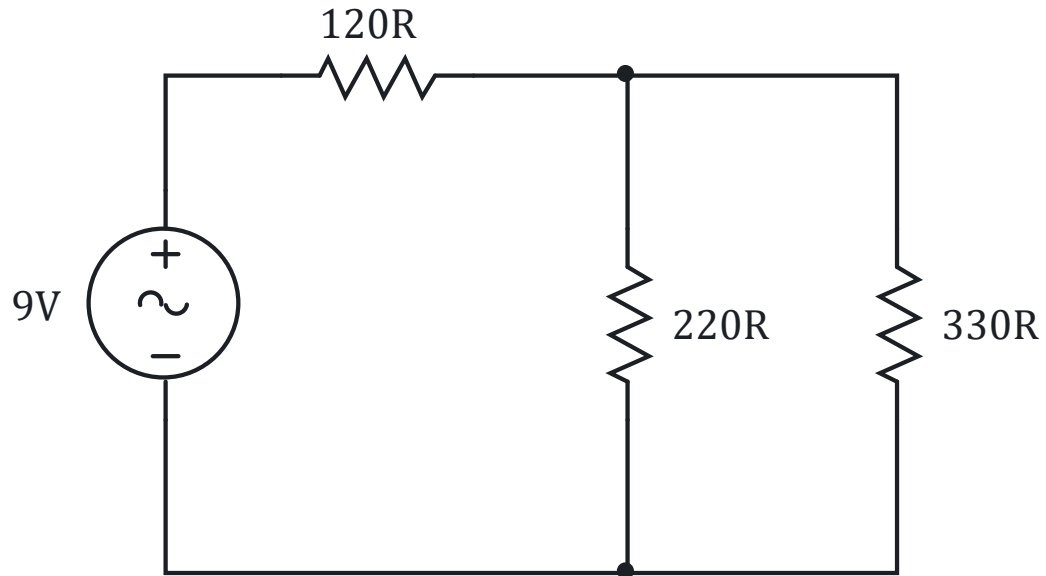
Solution



EXERCISE

Determine the current flowing through each resistor and the voltage drop across each resistor of the given circuit.

Solution



LABORATORY

